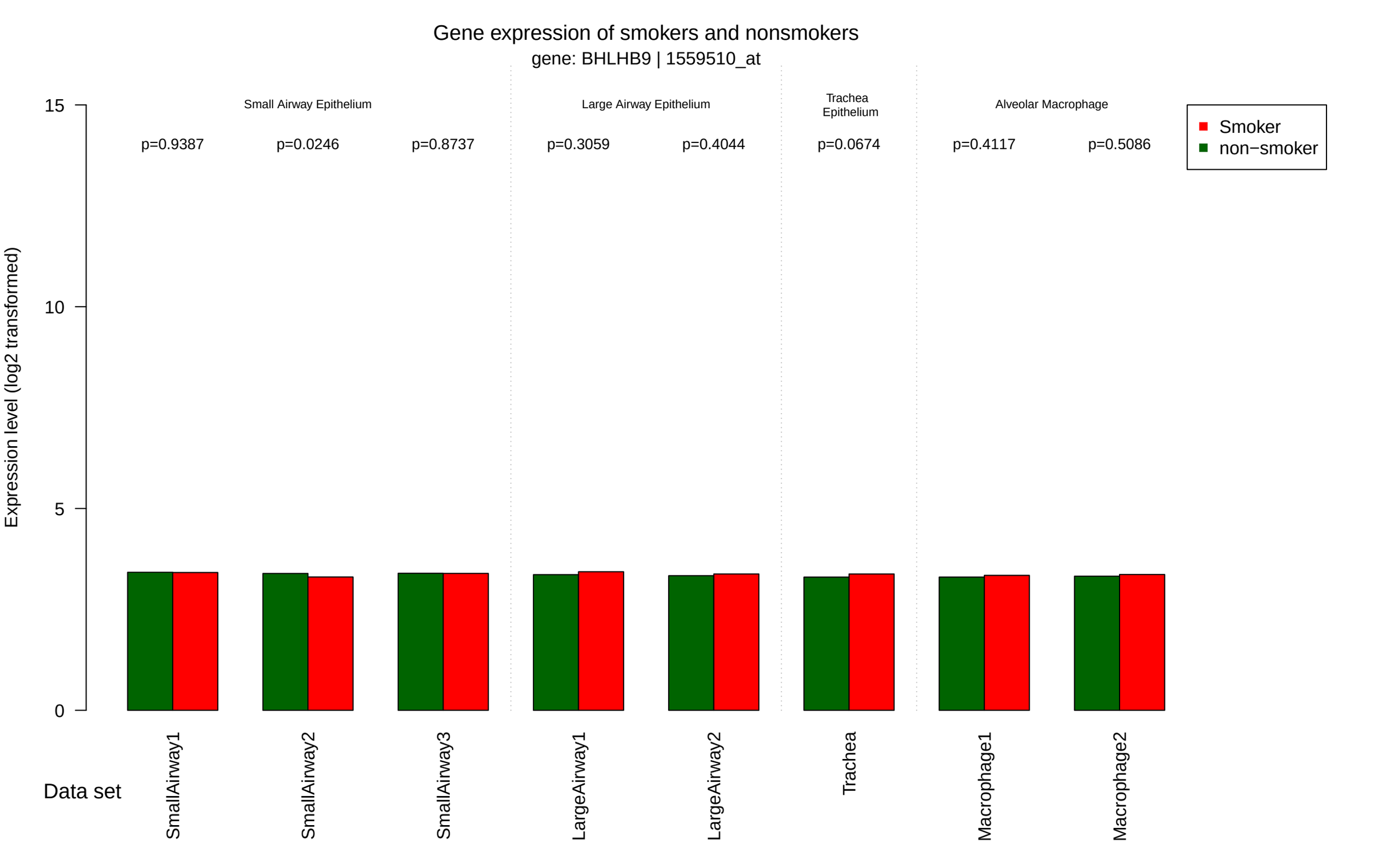
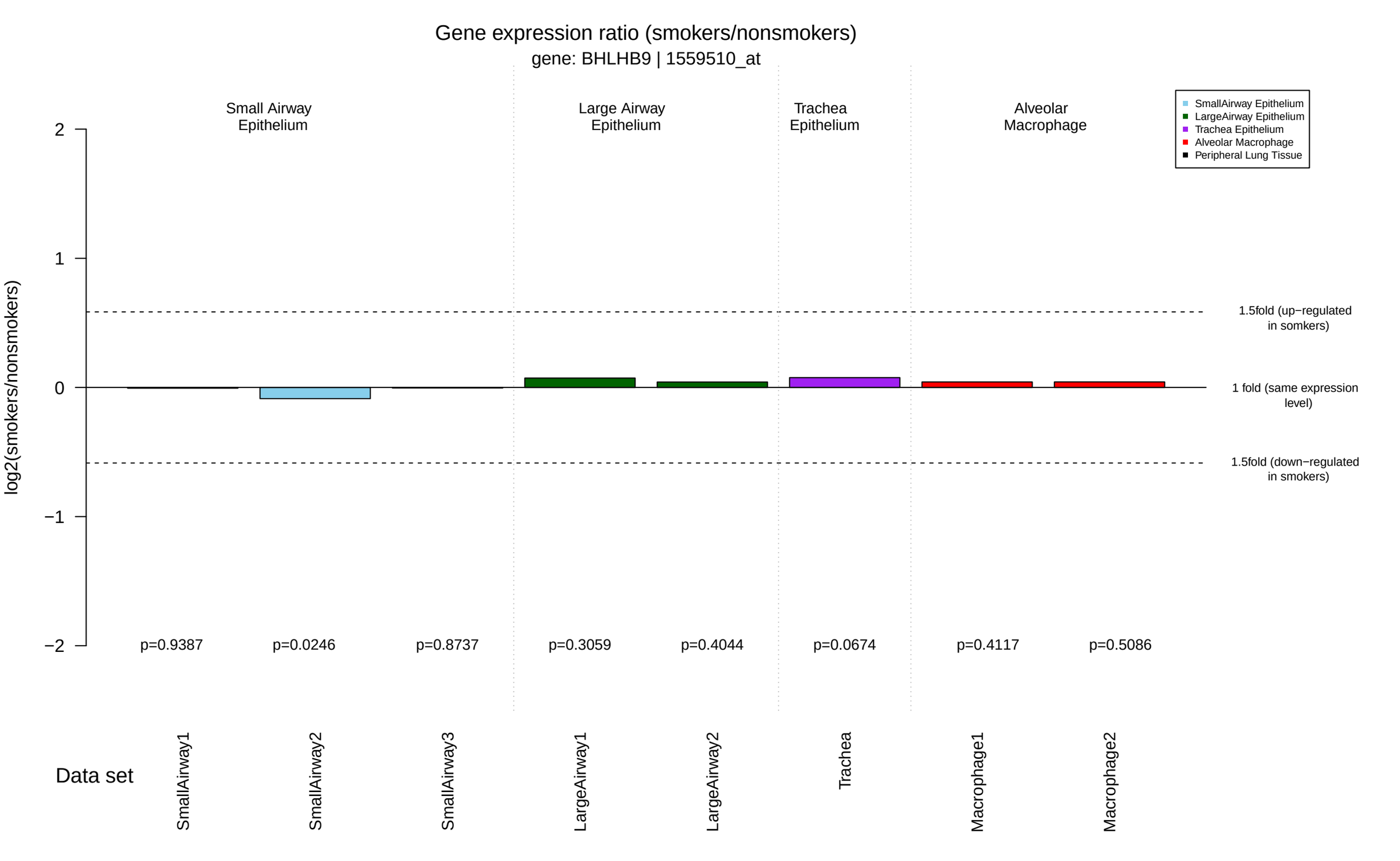
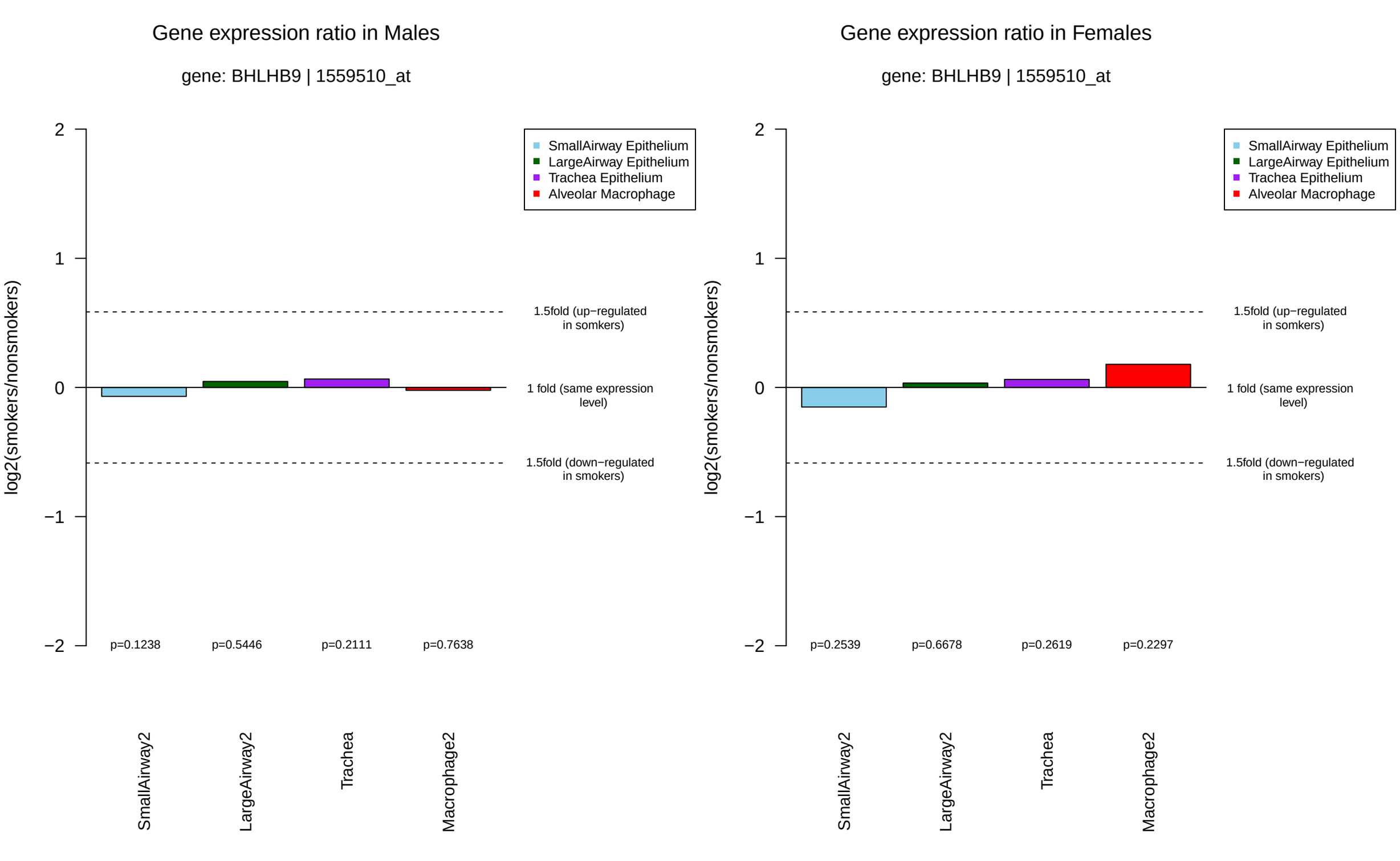
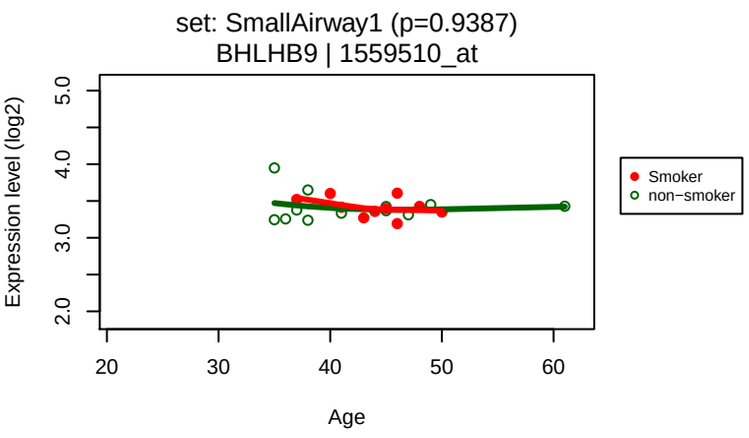
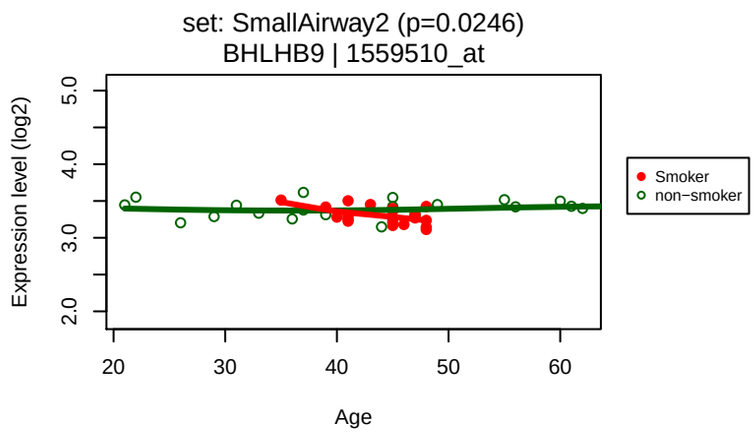
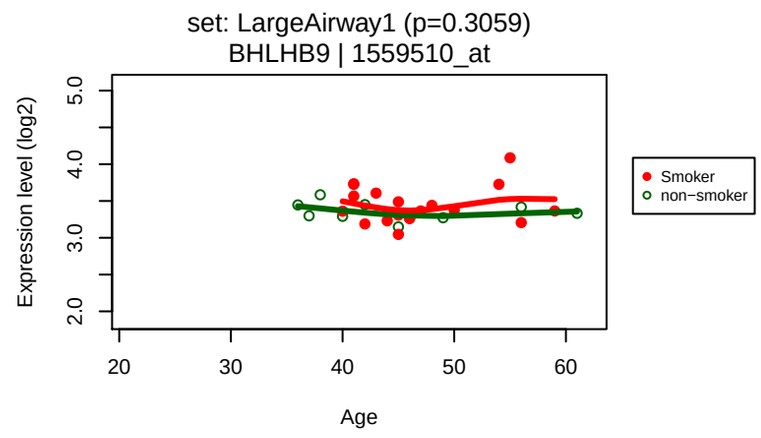

Gene expression of smokers and nonsmokers
gene: BHLHB9 | 1559510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9387
p=0.0246
p=0.8737
p=0.3059
p=0.4044
p=0.0674
p=0.4117
p=0.5086
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

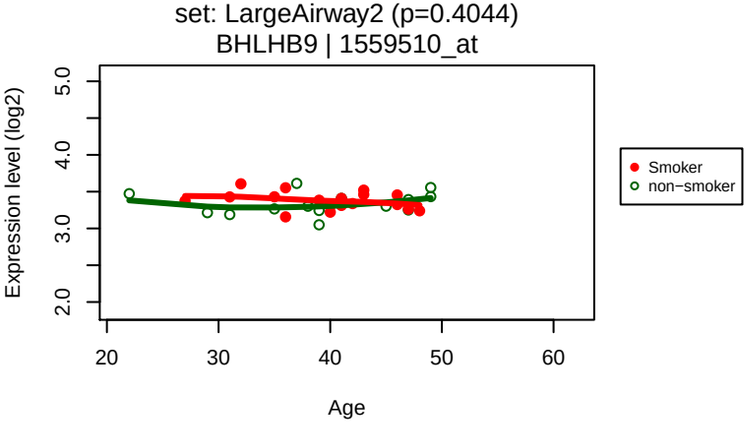

Gene expression ratio (smokers/nonsmokers)
gene: BHLHB9 | 1559510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9387
p=0.0246
p=0.8737
p=0.3059
p=0.4044
p=0.0674
p=0.4117
p=0.5086
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

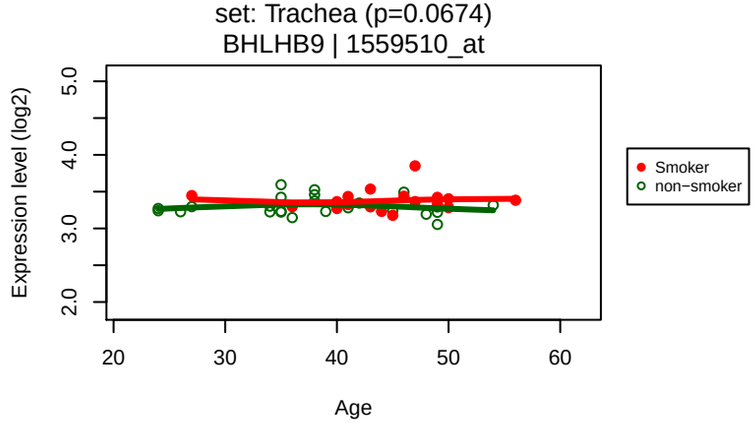

Gene expression ratio in Males
gene: BHLHB9 | 1559510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1238
p=0.5446
p=0.2111
p=0.7638
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHB9 | 1559510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2539
p=0.6678
p=0.2619
p=0.2297
SmallAirway2
LargeAirway2
Trachea
Macrophage2

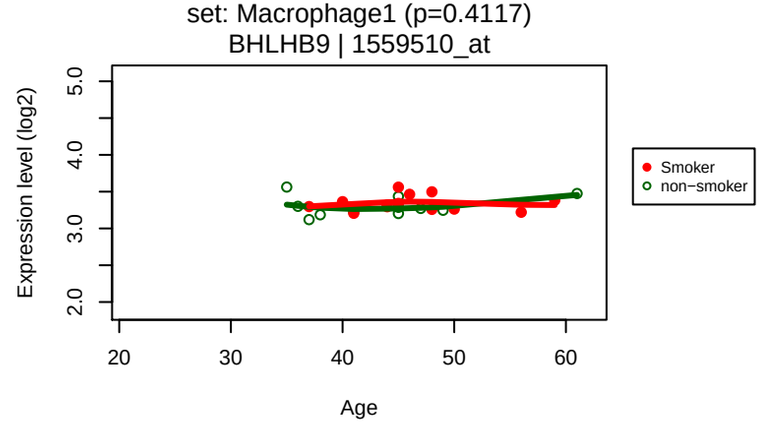

set: SmallAirway1 (p=0.9387)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker

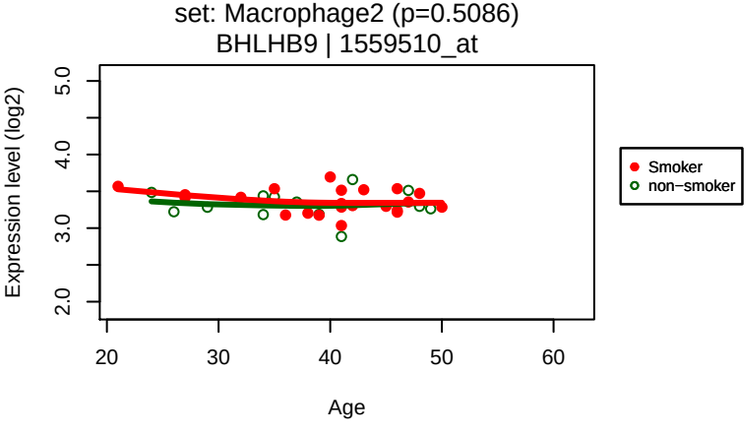

set: SmallAirway2 (p=0.0246)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker

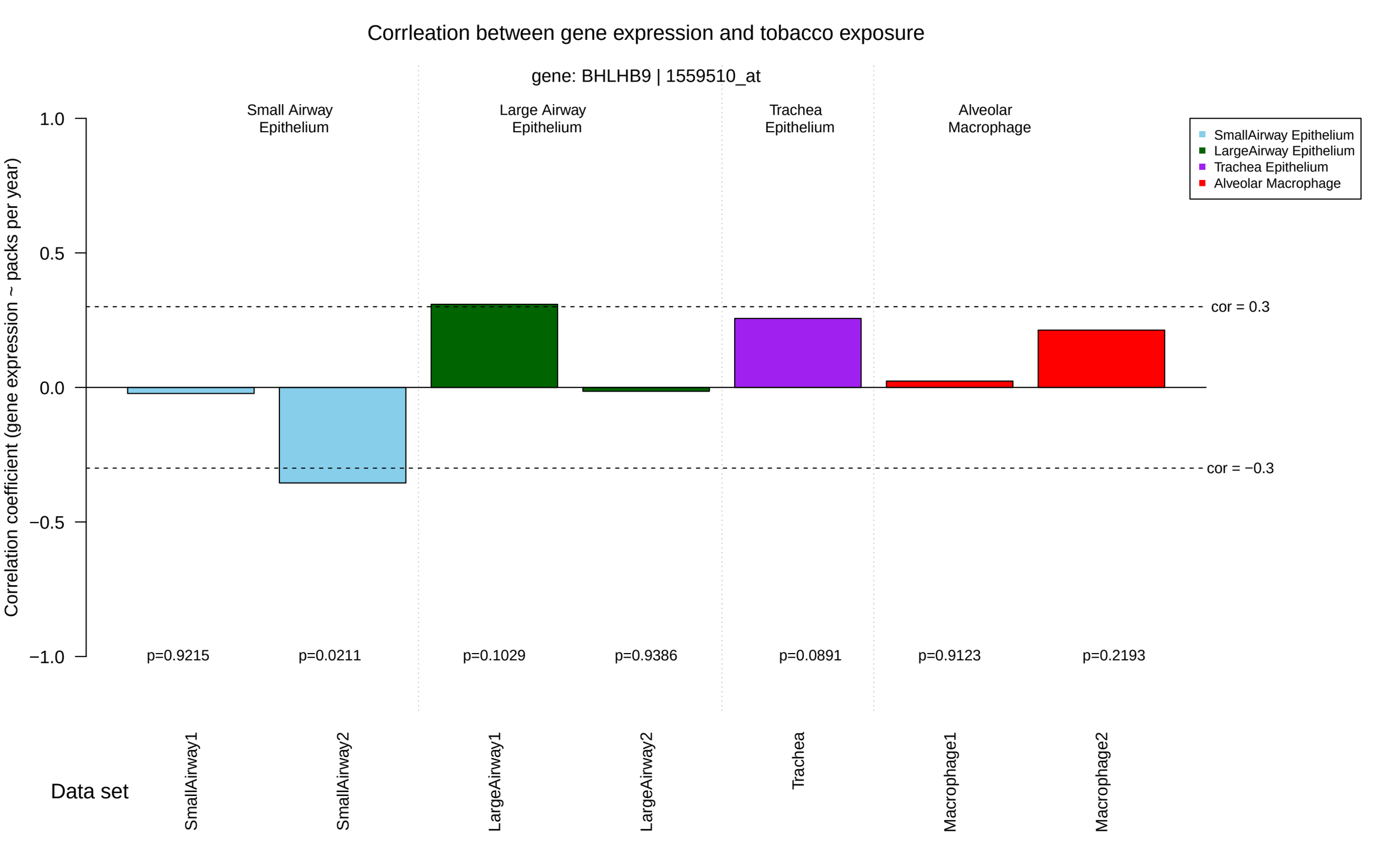

set: LargeAirway1 (p=0.3059)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4044)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0674)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4117)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5086)
BHLHB9 | 1559510_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHB9 | 1559510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9215
p=0.0211
p=0.1029
p=0.9386
p=0.0891
p=0.9123
p=0.2193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2